

Projects

Ranked by evidence, not hype.

ML CAPSTONE **PYTHON** **SCIKIT-LEARN** **DUCKDB**

FlyRank: Content Refresh Opportunity Scoring

A ranked queue that helps a content team with 20-50 pages per week of review capacity pick the right ones first. Built on the FlyRank starter dataset (30K rows) with client-holdout validation, transparent baseline comparison, and human-readable reason codes.

Key result: Random Forest Precision@50 = 0.74 vs. baseline 0.24 - a 3x lift.

[Read case study](#)

[View repo](#)

Supporting work

AI FLUENCY **PROMPT ENGINEERING**

The Prompt Ladder

Six versions of a single prompt, each adding one named technique. Cross-model comparison between Claude and ChatGPT with honest failure points.

Key finding: ChatGPT suggested PCA for a Random Forest - "sounds smart but wrong."

AI FLUENCY **PORTFOLIO**

Frame It as Cases

Case study framing for the FlyRank capstone using a voice card, three beats, and a before/after comparison of generic AI copy vs. edited voice.

AI FLUENCY **AUTOMATION**

Source-Grounded Research Notes

A 4-step no-code pipeline using NotebookLM + Claude to turn ML papers into structured research notes. 61% time reduction vs. manual note-taking.

AI FLUENCY **AGENT CONCEPTS**

Agent Concepts & MCP

Explainer on workflow vs. agent distinction, MCP primitives, and a concrete agent upgrade path for the research notes pipeline.

IN PROGRESS

ML Track: Weeks 3-7

Signal audit, baseline refinement, warehouse-scale training, validation, and capstone deployment.

